

Hackathon Guide

A2Tech360 Hackathon

Powered by IBM

A2Tech360 Hackathon participants build their AI-driven solutions using IBM's ecosystem of AI products and cloud services. This guide covers the five tools you have access to during the 48-hour event:

- **IBM Bob** — IBM's AI productivity assistant. Featured AI tool, available via free trial.
- **IBM watsonx.ai** — Foundation model studio with the Prompt Lab and programmatic API/SDK access.
- **IBM watsonx Orchestrate** — The agentic control plane for scaling and governing your AI
- **IBM Open Source** — IBM Granite models and the BeeAI agentic framework, runnable locally or self-hosted.
- **IBM Cloudant** — Fully managed JSON document database for application data and state.
- **IBM Code Engine** — Fully managed serverless platform for deploying containerized apps, jobs, and functions.

Use what serves your project. None of these are required individually, but each section below covers how to get started so your team can pick the right tools for the build.

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The hackathon challenge

In this hackathon, you will **build an AI-driven solution** for the following theme:

Make work less boring

This is your chance to show the world what “different” could look like when it comes to business. Build a proof-of-concept, AI-driven solution that brings creativity into how businesses operate. Whether it’s generating new ideas, designing creative content, predicting innovation or market trends, enhancing brainstorming processes, or nurturing creative strategy — identify an activity that can be improved and build a solution that empowers businesses to innovate faster and more creatively.

Refer to the [Make work less boring example use cases in the Appendix](#).

A note on data sets before you begin

Participants are required to bring their own datasets to build the solution aligning to your use case. As you collect data for your project, you’ll want to use best practices. Here are helpful tips:

- Teams are responsible for ensuring data is compliant.
- Data from public websites may be used, if the terms allow for commercial use, but please keep a list of the websites you use.
- Do not use data or assets containing company confidential data, or any other data without permission from the data owner. Teams are responsible for getting approval.
- Do not use any client data.
- Do not use any data containing personal information (PI).
- Do not use data obtained from social media.

Get started with IBM Bob

IBM Bob is IBM's AI productivity assistant and the featured AI tool for this hackathon. A2Tech360 Hackathon participants get free trial access to Bob, which lets your team build, prototype, and iterate quickly during the 48 hours — without standing up infrastructure or managing models yourself.

Free trial plan and Bobcoin limitation

Each interaction with Bob's AI capabilities consumes Bobcoins. The free trial plan comes with a limitation of **40 Bobcoins** for **30 days**. When Bobcoins are exhausted or the 30 days end, the account becomes inactive unless upgraded to a [paid plan](#).

Plan ahead: With only 40 Bobcoins available across 30 days, plan your Bob usage strategically for the 48 hours of the hackathon. Use Bob for high-impact moments — drafting key content, exploring an idea, or testing a workflow — rather than every small task. Coordinate within your team on who is signed in so you don't accidentally double-spend Bobcoins on the same prompts.

Step 1 — Subscribe for the trial plan

- Open the [Bob register for trial URL](#) in your system browser.
- Create an IBM account by filling out the form and selecting the **Next** button. If you already have an existing IBM account, you can proceed to log in and complete the authentication.
- Verify your email by entering the 7-digit code sent to your email and check the optional “IBM keeping you informed on products and services” box. Select the **Submit** button.
- An IBM and Bob account will be created and you will get a confirmation that your trial account is ready.
- You will also receive an email confirmation for subscribing to the trial plan.

Step 2 — Get started with your free trial and access the admin portal

After successfully subscribing for the trial plan, the Bob team will take about 5 minutes to get your account ready. Once it is ready, you will receive an email confirmation, and then you can download and get started using Bob.

Access your Bob admin portal to track your Bob usage and see account-related information.

Follow the instructions below to access your admin portal:

- Log in to your [IBM Bob account](#) using your free trial subscription email.
- Access the [admin portal](#). If you have multiple Bob accounts, please ensure you select the correct instance.
- You will see your trial subscription plan details, including remaining Bobcoins and days left on your trial.

Tips for working efficiently with Bob

- Front-load your highest-leverage prompts. Use early Bobcoins to validate your idea and shape your approach before you're deep into building.
- Track your Bobcoin usage in the admin portal so you don't run out at a critical moment in the build.
- Coordinate within your team on who is signed in to Bob, so you don't accidentally double-spend Bobcoins on duplicate prompts.
- Save useful prompts and outputs locally — your Bob trial expires after 30 days, so keep a record of what worked for your final submission and write-up.

Accessing and utilizing IBM watsonx products

To begin building your solution, explore the capabilities and resources for each IBM watsonx product enabled for this hackathon. Note, **watsonx Orchestrate** is a **required** product in this hackathon and any other product (watsonx.ai) are optional.

1. [IBM watsonx Orchestrate \(Optional\)](#)

IBM watsonx Orchestrate is an intuitive, AI-powered platform that you can use to create, configure, and deploy intelligent agents that can automate business tasks. Whether you're automating repetitive workflows or building complex multi-agent systems, the platform is designed to support users of all skill levels.

2. [IBM watsonx.ai \(Optional\)](#)

A powerful AI studio that enables you to experiment with foundation models like IBM Granite through the Prompt Lab. It can be used to enhance your agents with natural language understanding, intelligent decision-making, and dynamic response generation—serving as an inference engine to make your AI agents smarter and more interactive.

1. IBM watsonx Orchestrate

After successfully [joining the IBM Cloud account](#), you can access watsonx Orchestrate to begin working on the platform and building your solution.

With watsonx Orchestrate, you can:

- Build and deploy agents
- Automate tasks such as scheduling, data entry, and approvals
- Start small and scale up with advanced tools and integrations
- Explore the clean, guided UI that simplifies agent creation and management.

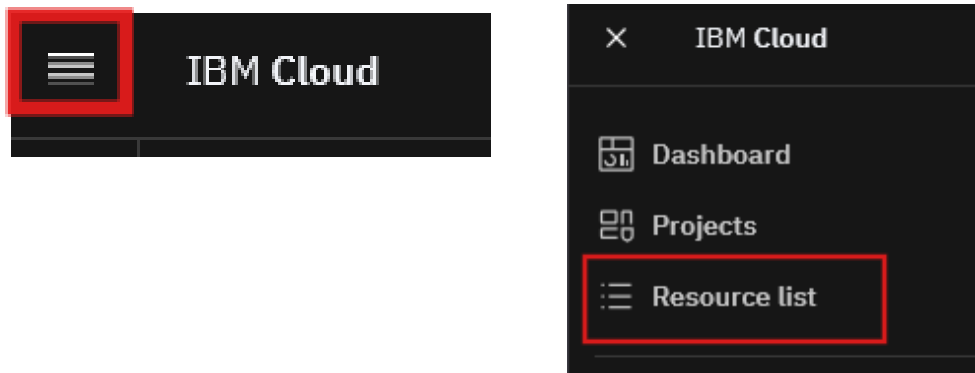
Explore a few [demos](#) on how to use watsonx Orchestrate.

These watsonx Orchestrate features/capabilities are out of scope for this hackathon:

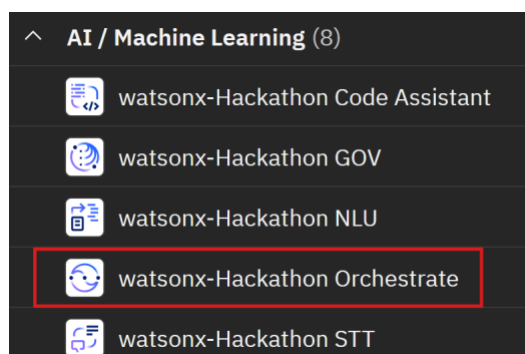
- Build with AI (Preview)

Accessing watsonx Orchestrate

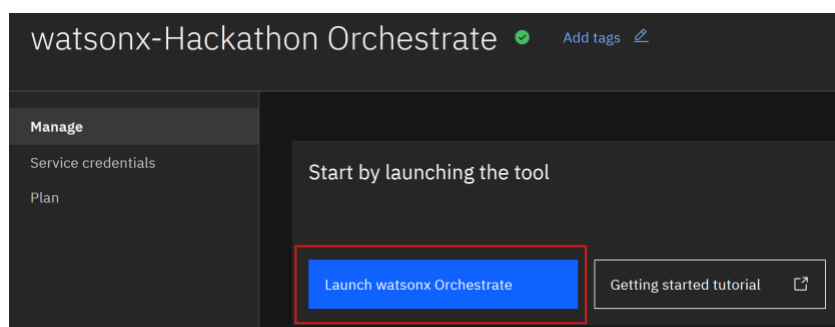
1. In your IBM Cloud account dashboard, select the **Navigation menu** on the top left of the dashboard and select the **Resource list** option.



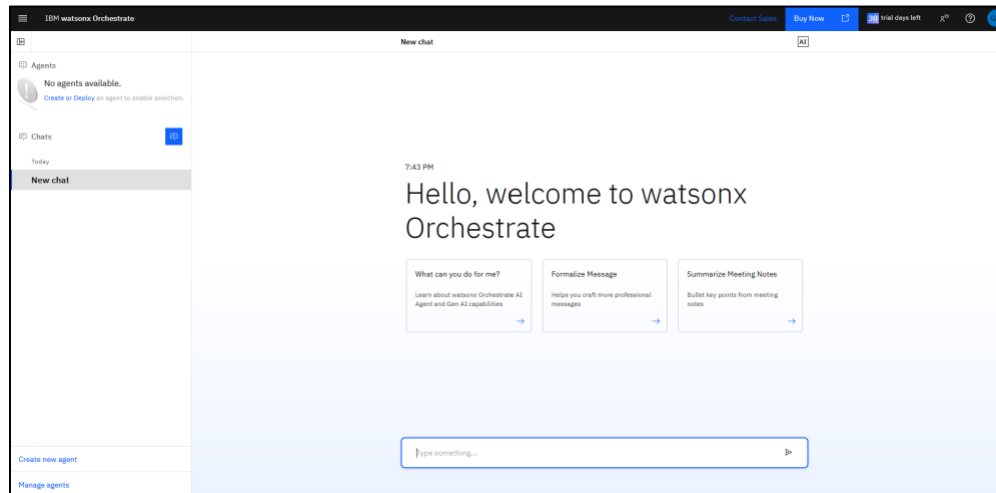
2. Expand the **AI / Machine Learning** section and select **watsonx-Hackathon Orchestrate** service.



3. You will be navigated to **watsonx-Hackathon Orchestrate** service instance dashboard. Select the **Launch watsonx Orchestrate** button.

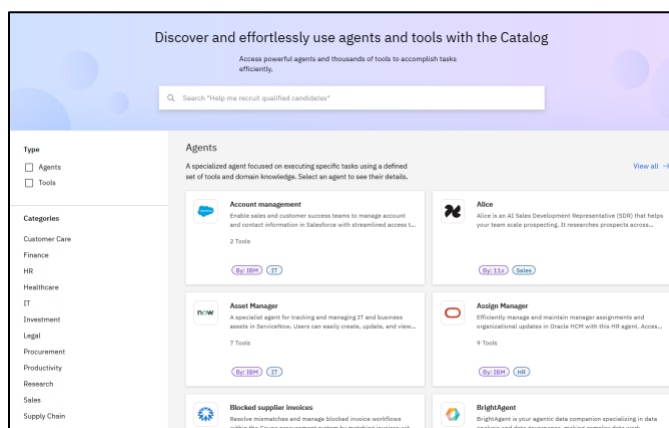


4. You will be navigated to the watsonx Orchestrate platform with a welcome message and a new chat window.



Discover the catalog

The IBM watsonx Orchestrate catalog is your gateway to a rich collection of prebuilt AI agents and tools, designed to support a wide range of business functions and use cases. Whether you're looking to automate tasks, enhance productivity, or integrate with backend systems, the catalog helps you find the right solutions quickly and efficiently. [Learn more about discovering the catalog.](#)



Building agents

In IBM watsonx Orchestrate, agents are a key component of the agentic AI framework, enabling you to create complex, dynamic systems that can adapt and respond to changing conditions.

By building agents, you can:

- Automate repetitive tasks
- Improve decision-making processes
- Enhance customer and employee experiences
- Increase operational efficiency

1. Prepare to build AI agents

Developing intelligent, scalable, and reliable agents in watsonx Orchestrate requires a strategic, multi-phase approach. It goes beyond writing prompts or connecting APIs. Successful agents need thoughtful planning, structured development, testing, and ongoing governance. The following overview breaks down each phase of agent development, highlighting key considerations. Learn more about [preparing to build AI agents](#).

2. Creating and customizing agents

Creating and customizing an AI agent involves defining its purpose and personality through thoughtful descriptions and selecting the most suitable foundation model that aligns with your brand or use case. You can also configure key elements like the welcome message and starter prompts to ensure the agent engages users effectively from the first interaction. Together, the configured components shape how the agent communicates, responds, and delivers value across conversations.

Learn [how to create an agent from scratch or from a template](#).

The screenshot shows a 'Create an agent' dialog box with two main options: 'Create from scratch' (selected with a radio button) and 'Create from template' (unselected). Below these options are input fields for 'Name' (with placeholder text 'Name your agent') and 'Description' (with placeholder text 'Example: This agent helps answer customer questions about return and cancellation policies.'). A small icon of a document with a pencil is next to the description field. Below the description field is a link that says 'What makes a good description?'. At the bottom left is a 'Cancel' button and at the bottom right is a 'Create' button.

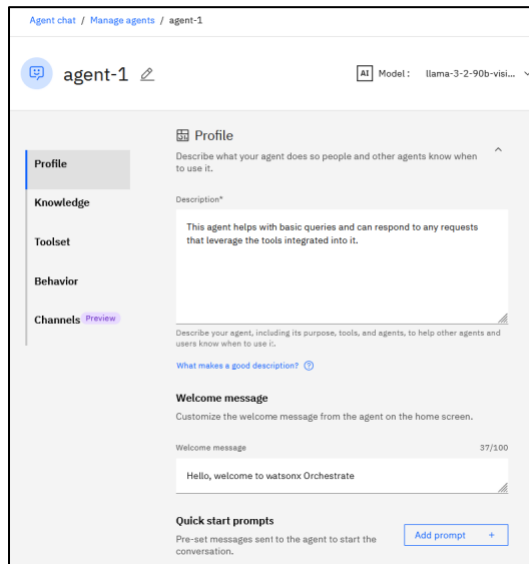
Expanding your agent's capabilities:

To enhance your agent's abilities, explore the following configuration options:

- **Define a profile**

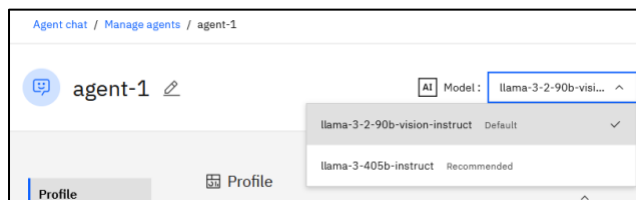
Provide a clear and specific description of the agent's purpose. This helps in multi-agent orchestration by enabling accurate selection based on capabilities.

See [Defining the description of your agent](#).



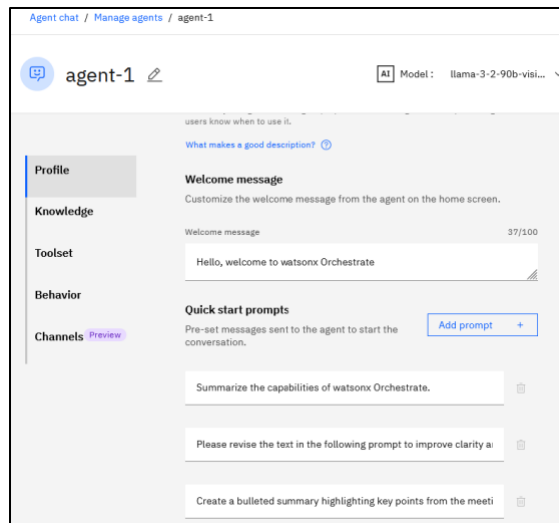
- **Choose an AI model**

Choosing the right AI model helps your agent understand user intent, reason through tasks, and generate reliable responses. Your choice directly affects accuracy, performance, and cost. See [Choosing an AI model](#).



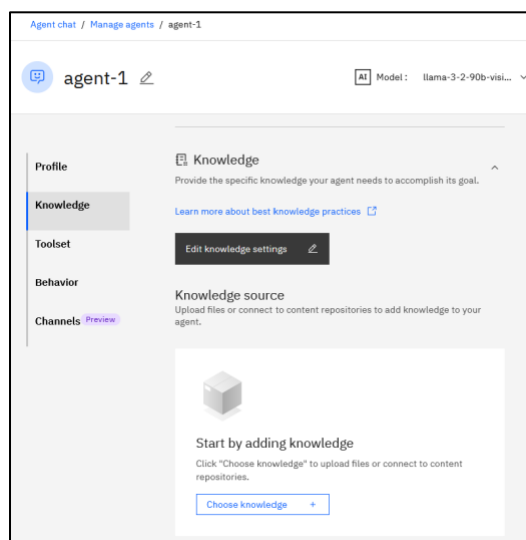
- **Customize the welcome message and starter prompts**

Customize the welcome message and starter prompts to guide users when they begin interacting with the agent. See [Customizing the welcome message and starter prompts](#).



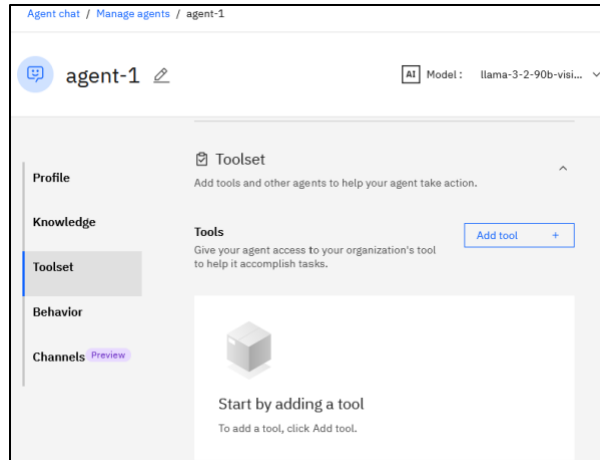
- **Add knowledge**

Enhance the agent's domain expertise by adding contextual knowledge from files or content repositories. See [Adding knowledge to agents](#).



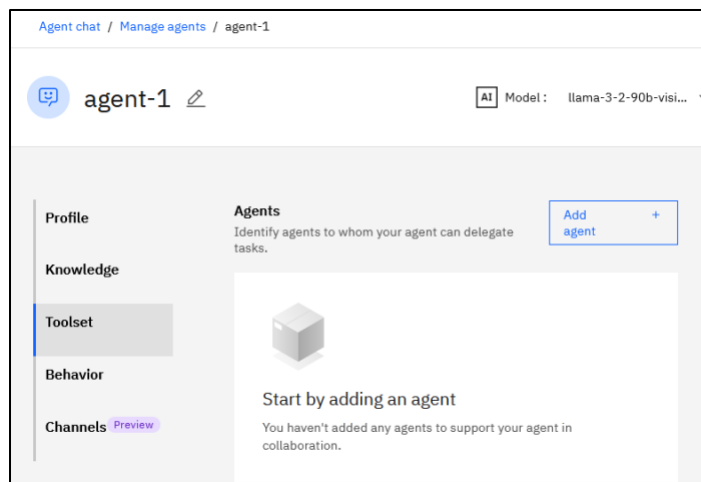
- **Add and manage tools**

Integrate tools to enable the agent to perform automated tasks such as retrieving data or sending emails. See [Adding and managing tools for agents](#).



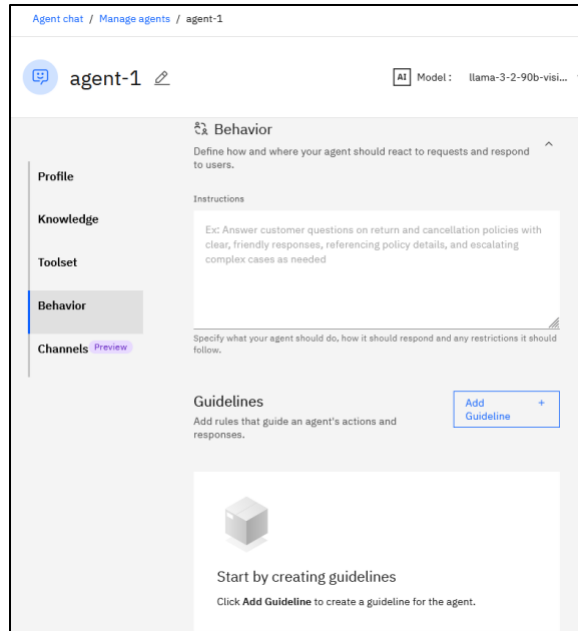
- **Orchestrate agents**

Enable multi-agent orchestration by adding collaborator agents that work together to achieve shared goals. See [Orchestrating agents](#).



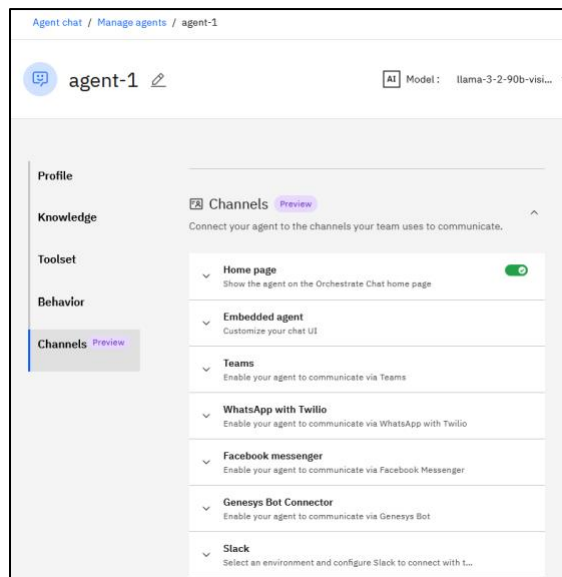
- **Add instructions**

AI agents operate based on a set of behavioral instructions that define how they respond to user inputs, perform tasks, and interact across different channels. These instructions serve as the agent's internal guidelines, helping it act consistently and effectively in various scenarios. See [Adding instructions to agents](#).



- **Connect agents to channels**

Channels are the interfaces through which you can interact with a conversational agent, such as messaging platforms and voice assistants. Integrate your agent with these channels to access services and support directly within your preferred environments to enable seamless and scalable communication. See [Connecting agents to channels](#).



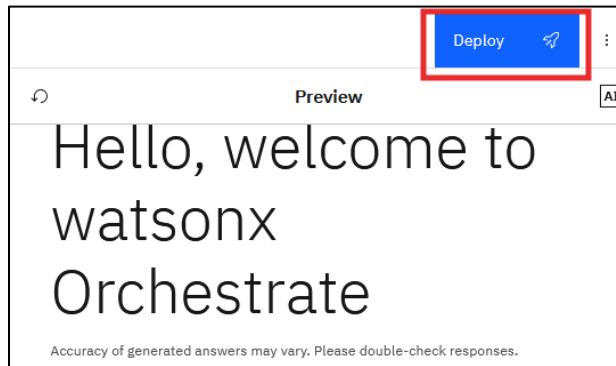
- **Configure rich responses**

Enable rich responses to incorporate multimedia and structured elements, making

AI interactions clearer, more efficient, and engaging for users. See [Configuring rich responses from the AI assistant builder](#).

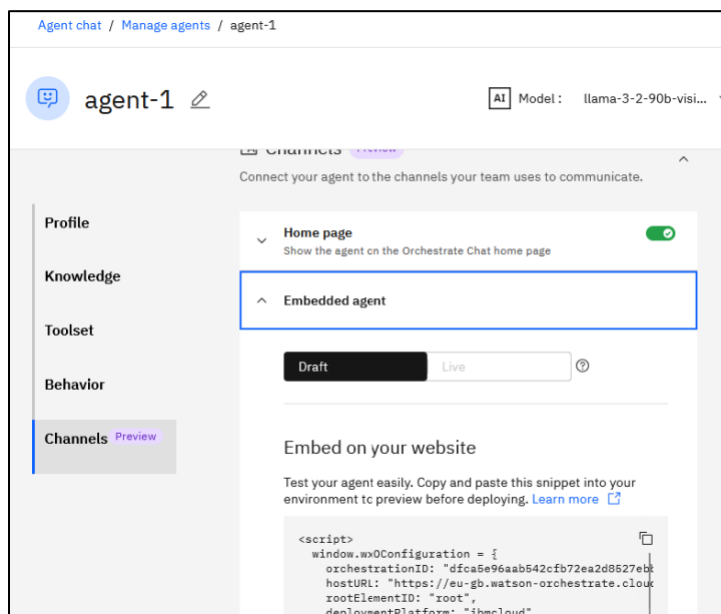
- **Deploy agents**

Finalize the setup by deploying the agent to make it available in live environments such as chat interfaces. See [Deploying agents](#).



- **Embed agents in applications**

IBM watsonx Orchestrate makes it easy to embed intelligent agents directly into your web applications by using the embedded chat feature. It enables interactive, contextual conversations while maintaining enterprise-grade security and flexibility. See [Embedding agents in applications](#).



- **Building agents using the ADK**

You can build powerful, customizable agents using the IBM watsonx Orchestrate Agent Development Kit (ADK). [Learn more about using the ADK.](#)

3. Building tools

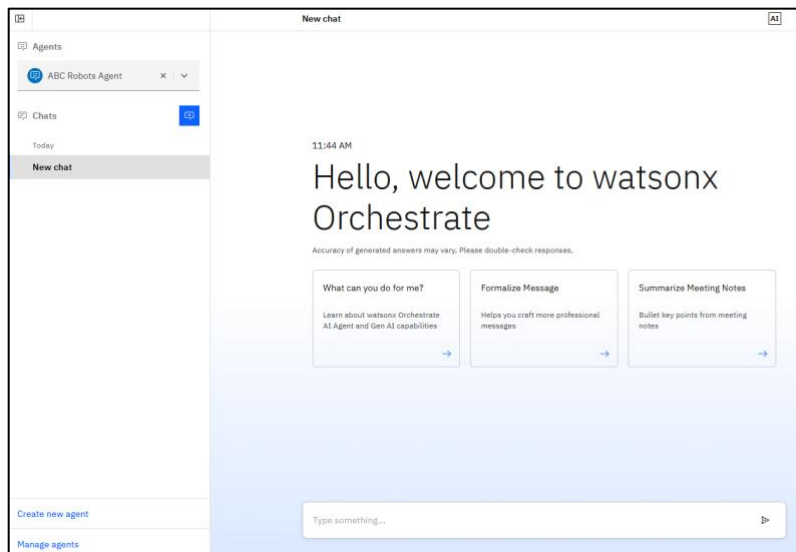
Tools act as interfaces to external capabilities, enabling agents to interact with systems, retrieve data, run calculations, or trigger workflows that go beyond their built-in reasoning. You can build new tools and edit existing tools for your agents to use.

Following are the different ways to add a tool:

- [Creating agentic workflows](#)
- [Adding tools to an agent](#)
 - [Catalog](#)
 - [Local instance](#)
- [Importing external tools](#)
 - [MCP server](#)
 - [OpenAPI](#)

Using agents in the chat

In IBM watsonx Orchestrate, agents collaborate to automate tasks and manage workflows. [Learn more using agents in Orchestrate Chat.](#)



Managing app connections

To use the external applications within IBM watsonx Orchestrator, you must establish a connection between them which acts as bridge enabling communication between watsonx Orchestrator and the external applications. [Learn more about managing app connections and credentials](#).

watsonx Orchestrator API

IBM watsonx Orchestrator provides a set of APIs to boost your experience using the features from the product.

- [Getting started with the API](#)
Begin using the API with practical, step-by-step guidance.
- [API reference](#)
Get the methods that you can use to call custom projects, or other resources from the AI chat.

Building AI assistants

In IBM watsonx Orchestrator, you build the AI assistant by using AI assistant builder. AI assistant builder is a chat interface builder that helps to deploy an engaging and embedded chatbot experience. AI assistant builder integrates the power of large language models (LLMs)

and conversational capabilities of watsonx Assistant to enable responsive and interactive conversation between the users and watsonx Orchestrate.

To learn more, see [Building AI assistants in AI assistant builder](#).

Quick start hands-on exercises

Try the quick start hands-on exercises for sample use cases to get started with using watsonx Orchestrate:

Important notes:

- You must use the hackathon-provisioned IBM Cloud account to access and use watsonx Orchestrate platform to try the below sample exercises.

watsonx Orchestrate SaaS sample exercises:

- [Develop agents with no code using watsonx Orchestrate](#)
A hands-on guide for developing a multi-agent system in watsonx Orchestrate with no code that can communicate with the agents on various channels.
- [Creating intelligent, reusable agentic workflows on watsonx Orchestrate with no code](#)
A hands-on guide for creating AI-driven, human-in-the-loop workflows without writing code.

- [Build context-aware AI agents with watsonx Orchestrate and Astra DB](#)
AI agents are only as effective as their data, making precise retrieval essential for reliable results.
- [AgentOps in watsonx Orchestrate: Observability for Agents with Langfuse and IBM Telemetry](#)
A developer's guide for instrumenting and observing Agents with Langfuse and IBM Telemetry in watsonx Orchestrate.
- [Building and deploying prebuilt domain agents in watsonx Orchestrate](#)
Explore how to work with the prebuilt domain agents and integrate them into existing workflows.
- [Build a scheduled agentic workflow for daily notifications](#)
A practical example of building, importing, and scheduling an agentic workflow that automatically sends daily email notifications using watsonx Orchestrate.
- [Multi-agent orchestration with watsonx Orchestrate](#)
A hands-on journey to design, extend, and integrate AI agents by using Langflow, IBM Granite models, Astra DB, the AI Gateway, and a custom user interface on watsonx Orchestrate.

watsonx Orchestrate Agent Development Kit (ADK) sample exercises:

- [Getting Started with watsonx Orchestrate Agent Development Kit](#)
IBM watsonx Orchestrate includes the Agent Development Kit (ADK) which is a set of developer-focused tools for building, testing, and managing agents. With the ADK, developers get the freedom and control to design powerful agents using a lightweight framework and a simple CLI. You can define agents in clear YAML or JSON files, create custom Python tools, and manage the entire agent lifecycle with just a few commands.
- [Build an AI agent with Langflow, Granite 4.0 models, and watsonx Orchestrate](#)
A hands-on guide for creating Langflow tools, consuming them with watsonx Orchestrate Agent Development Kit, and using IBM Granite 4.0 Micro model for inference.
- [Connecting to MCP tools with watsonx Orchestrate](#)

A hands-on guide for creating MCP tools and consuming them with watsonx Orchestrate Agent Development Kit.

- [Extend Your AI Agents with External LLMs Using watsonx Orchestrate and AI Gateway](#)

A hands-on guide to bring your own model (BYOM) into watsonx Orchestrate using AI Gateway.

- [Create consistent evaluation workflows for AI agents](#)

A hands-on guide to setup and use a structured evaluation strategy to test, benchmark, and improve AI agent performance.

Get started with IBM watsonx.ai

IBM watsonx.ai is a foundation model studio that supports prompting, programmatic API/SDK access, and AI agent development. Through the Prompt Lab, you can experiment with IBM Granite and other foundation models. watsonx.ai can also serve as the inference layer for AI agents your team builds during the hackathon.

To access watsonx.ai, participants must be registered for the hackathon and have access to the hackathon site. Once registered, follow the “Complete the hackathon” page instructions to request a pre-configured IBM Cloud account for your team.

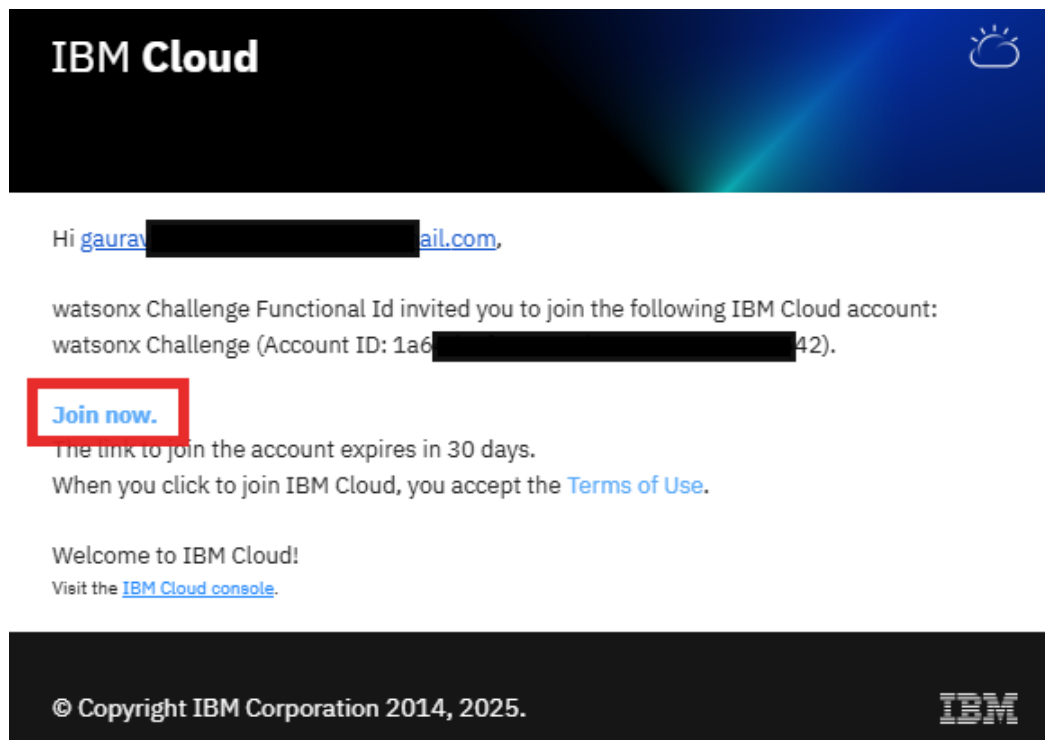
watsonx.ai credit usage: Each provisioned account receives **\$100 in credits (for both watsonx.ai and Code Engine)**. Foundation model inferencing consumes tokens (1,000 tokens = 1 RU at \$0.0001 USD), and notebook runtimes are billed per Capacity Unit Hour (CUH) at \$1.02 USD/CUH. Email alerts are sent at 25%, 50%, and 80% usage; accounts are deactivated within an hour at 100%. Plan accordingly and back up your work.

Out of scope features: Agent Studio (Beta), Deploy on IBM Cloud/watsonx.ai, Bring your own model, Fine-tuning, and Cloud Object Storage are not available for this hackathon.

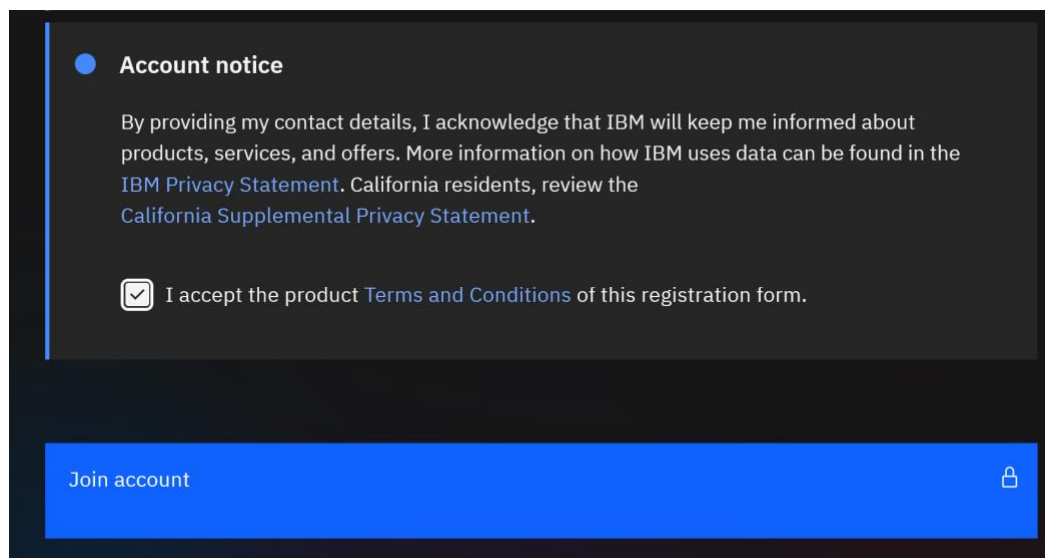
Access your IBM Cloud account

Once provisioned, all team members will receive an email invite to join the cloud account. Follow these steps:

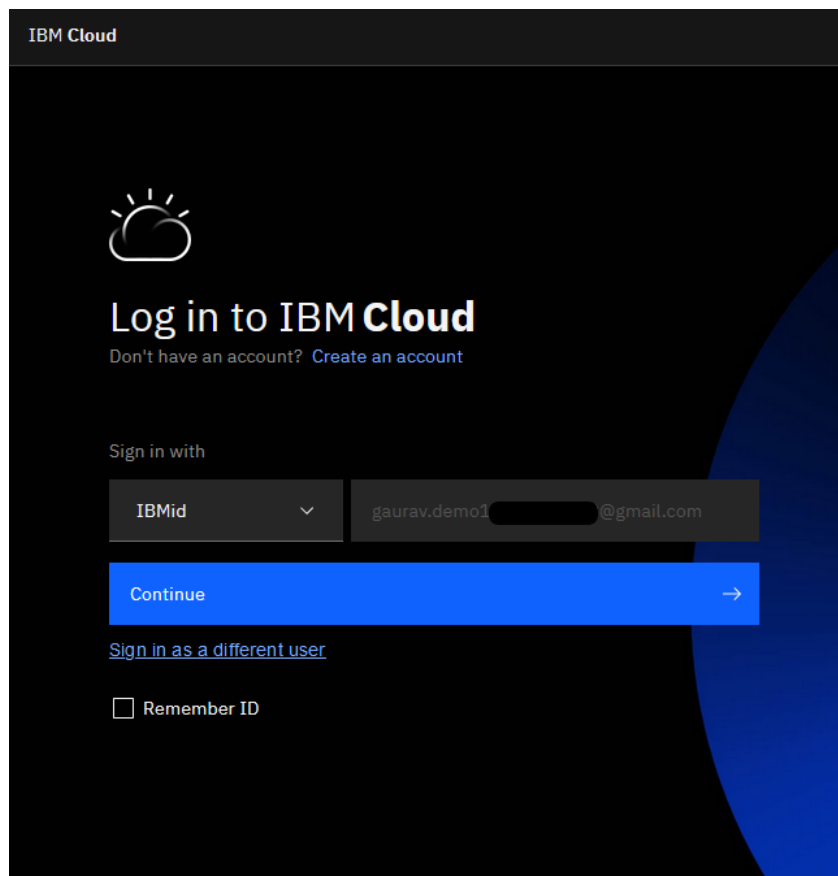
- Check the email inbox you used to register for the hackathon and open the email from the IBM Cloud team.
- Click the **Join now** link in the email. A new browser tab will open with the cloud account sign-up page.



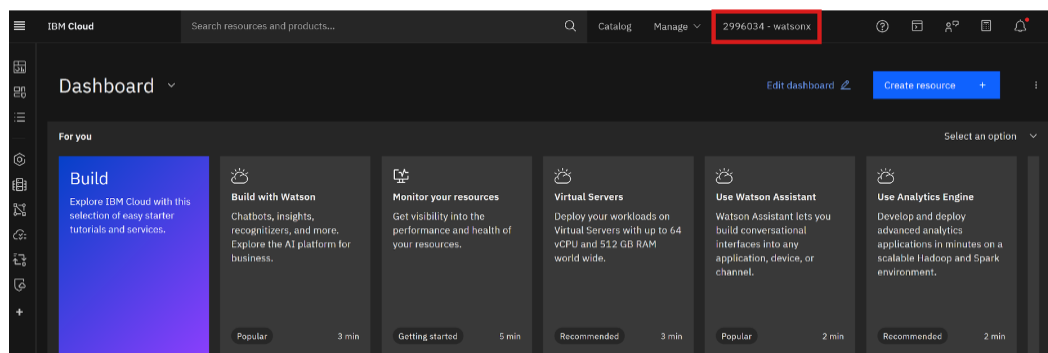
- Review your account and personal information.
- Read and accept the Account notice and click the **Join Account** button.



- Complete the authentication process by clicking the Continue button.

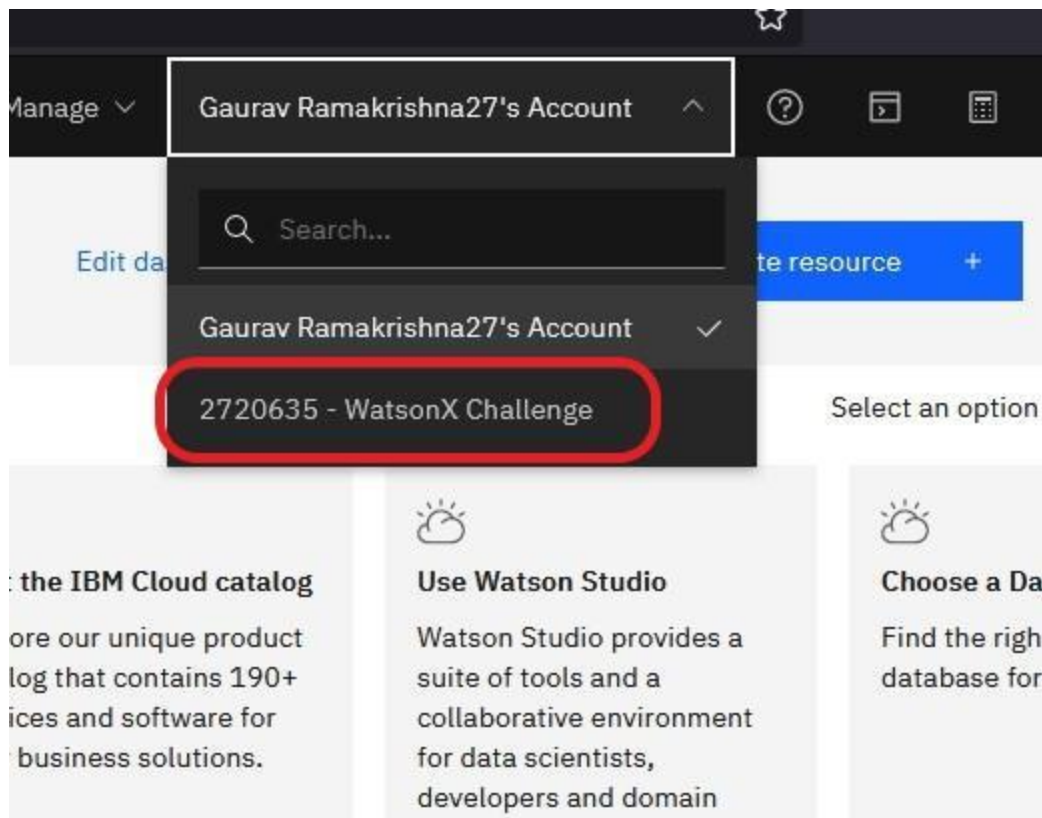


- After you authenticate successfully, you will be taken to the IBM Cloud dashboard.



Important note

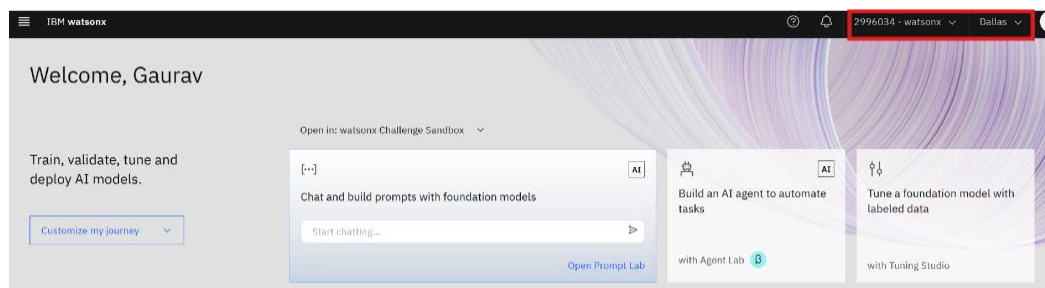
If you have an existing personal IBM Cloud account on the same email/IBMid, you may sometimes be directed to your personal account. In this case, switch to the **xxxxxxx – watsonx** account from the account drop-down at the top-right of the dashboard.



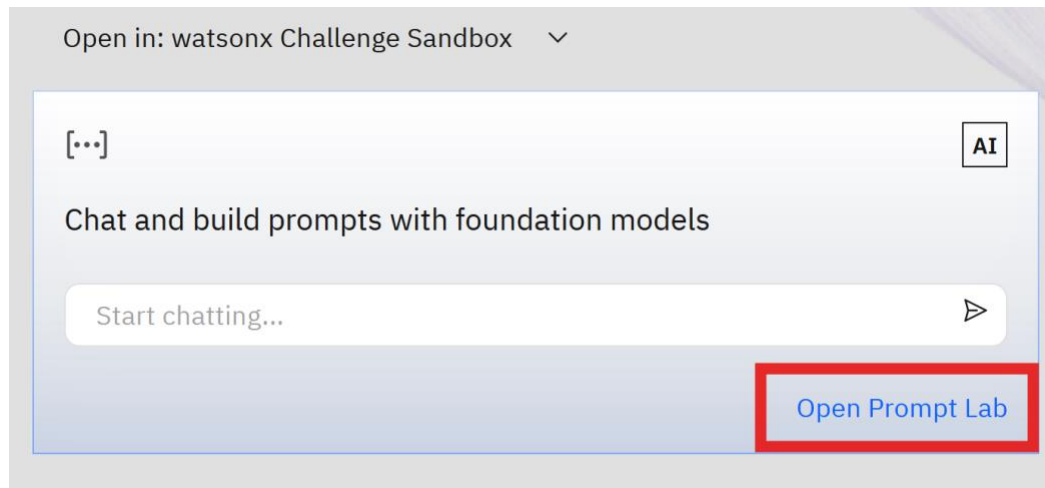
Access Prompt Lab on watsonx.ai

After joining the IBM Cloud account, you can access the Prompt Lab to work with foundation models supported on the platform.

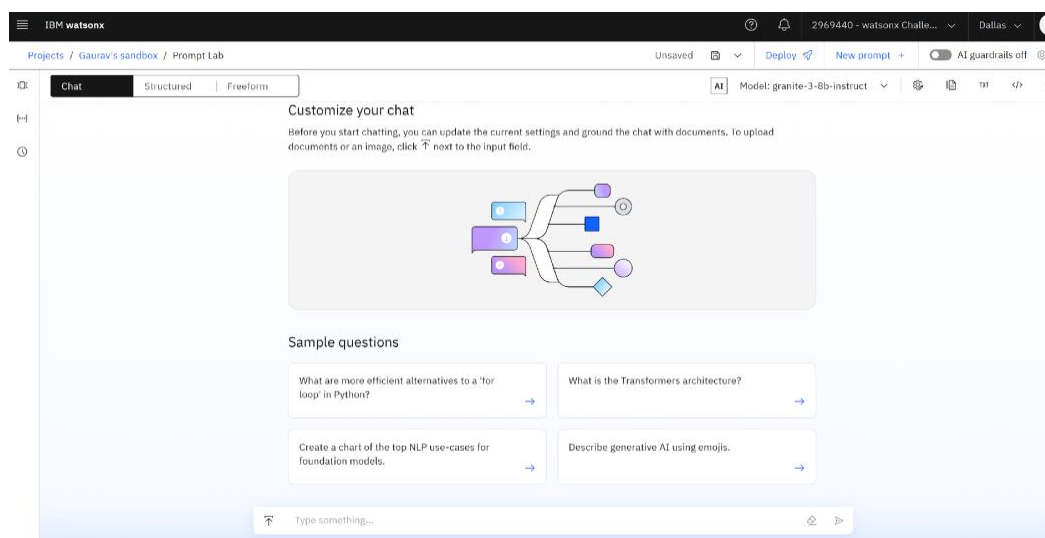
- Log in at <https://dataplatform.cloud.ibm.com/wx/home?context=wx> with the email you used for IBM Cloud.
- Confirm the account name is xxxxxxxx – watsonx and the region is set to Dallas.



- Click **Open Prompt Lab** on the “Chat and build prompts with foundational models” widget.



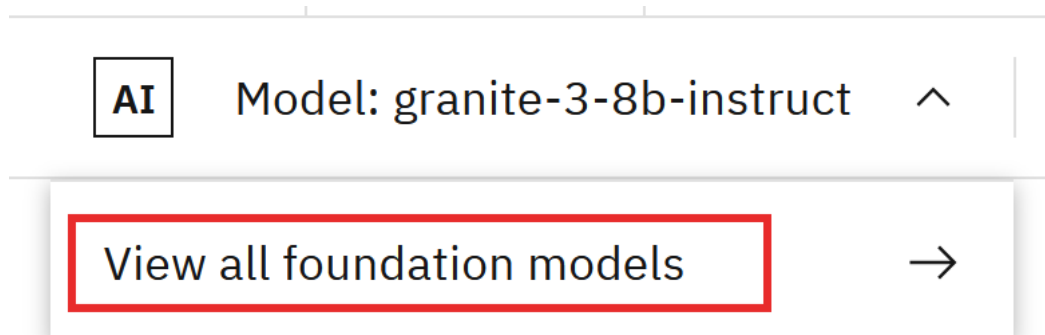
- The Prompt Lab editor opens with a chat window to get you started.



Selecting an IBM Granite model

A **granite-3-8b-instruct** model is pre-selected. To switch:

- Open the AI Model drop-down and click **View all foundation models**.



- Clear all filters and search “granite”. All Granite series models will be displayed.

Select a foundation model

To choose a model, review characteristics such as tasks that models perform. Compare model benchmarks with scores in the range 0–100. Higher scores are better.

All models		Model benchmarks			
granite Want to bring your own model?					
granite-3-8b-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-13b-instruct-v2 The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-20b-code-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-20b-multilingual The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-3-2-8b-instruct Granite 3.2 8b Instruct is a text-only model capable of reasoning which can be enabled or disabled by the user to use the right... Provider: IBM Type: Provided model	granite-3-2b-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model
granite-34b-code-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-3b-code-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-8b-code-instruct The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-guardian-3-2b The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-guardian-3-8b The Granite model series is a family of IBM-trained, dense decoder-only models, which are particularly well-suited for generative... Provider: IBM Type: Provided model	granite-vision-3-2-2b Granite 3.2 Vision is a image-text-in, text-out model capable of understanding images like charts for enterprise use cases for compute... Provider: IBM Type: Provided model

See [choosing a model](#) for guidance on modalities, supported languages, tuning, and indemnification.

Programmatic access (API endpoint)

To call watsonx.ai APIs/SDKs, you’ll need three values: a **project ID**, **endpoint URL**, and **API key**.

- On the [watsonx.ai home page](#), scroll down to the **Developer access** section.

Developer access ⓘ

Project or space

Project ID

Project or space

00000000-0000-0000-0000-000000000000

watsonx.ai URL

https://us-south.ml.cloud.ibm.com

Used to call watsonx.ai APIs such as LLM inferencing, embedding, training, and chatting.

Create API key +

Manage IBM Cloud API keys →

- Select **Project or space** → **watsonx Challenge Sandbox**. A project ID will be displayed.

Project or space

Project ID

watsonx Challenge Sandbox

0508[REDACTED]ca0

- A default watsonx.ai endpoint URL will be displayed for the Dallas region.

watsonx.ai URL

<https://us-south.ml.cloud.ibm.com>



Used to call watsonx.ai APIs such as LLM inferencing, embedding, training, and chatting.

- Click **Create API key**, enter a name, choose “Disable the leaked key”, and click **Create**.

Create API key



You need an IBM Cloud API key to create the IAM bearer token for authenticating your API requests. [Learn more](#)

Name

watsonx API key

Description (optional)

Enter name

Choose what to do if this key is leaked:

- ☒ Disable the leaked key
☐ Delete the leaked key
☐ Nothing

Cancel

Create

- Copy and store the API key securely — you can also download the file.

API key successfully created



Copy the API key or click download to save it. You won't be able to see this API key again, so you can't retrieve it later. The API key is no longer displayed after (292) seconds.

API key

.....



Copy



Download



Access the prompt code from Prompt Lab

To prompt an AI model programmatically, click the **View code** icon at the top-right of the Prompt Lab editor.

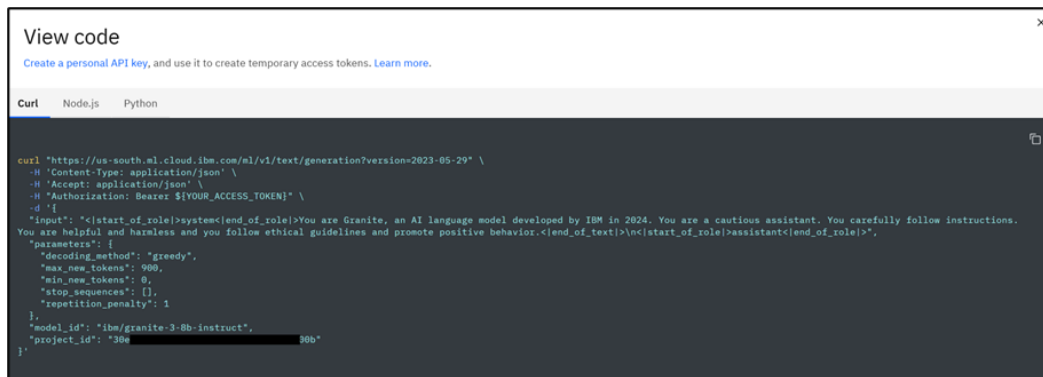


TXT

</>



The prompt code is available as cURL, Node.js, and Python.



```
View code
Create a personal API key, and use it to create temporary access tokens. Learn more.

Curl Node.js Python

curl "https://us-south.ml.cloud.ibm.com/ml/v1/text/generation?version=2023-05-29" \
-H 'Content-Type: application/json' \
-H 'Accept: application/json' \
-H 'Authorization: Bearer ${YOUR_ACCESS_TOKEN}' \
-d '{
  "input": "<start_of_role>system<end_of_role>You are Granite, an AI language model developed by IBM in 2024. You are a cautious assistant. You carefully follow instructions. You are helpful and harmless and you follow ethical guidelines and promote positive behavior.<end_of_text>\n<start_of_role>assistant<end_of_role>",
  "parameters": {
    "decoding_method": "greedy",
    "max_new_tokens": 980,
    "min_new_tokens": 0,
    "stop_sequences": [],
    "repetition_penalty": 1
  }
},
{
  "model_id": "ibm/granite-3.0b-instruct",
  "project_id": "30a"
}]'
```

To authorize requests, replace the **\${YOUR_ACCESS_TOKEN}** placeholder with an IAM access token generated from your API key.

Generating an IAM access token

POST to **<https://iam.cloud.ibm.com/identity/token>** with header **Content-Type: application/x-www-form-urlencoded** and body **grant_type=urn:ibm:params:oauth:grant-type:apikey&apikey=YOUR_API_KEY**. IAM tokens are valid for up to 60 minutes — check the *expires_in* property and refresh as needed.

watsonx.ai API capabilities

Available capabilities to explore:

- [Chat](#)
- [Tool calling](#)
- [Text generation](#)
- [Time series](#)
- [Text rerank](#)
- [Embeddings](#)
- [Text extraction](#)

See [supported API functionality by model](#).

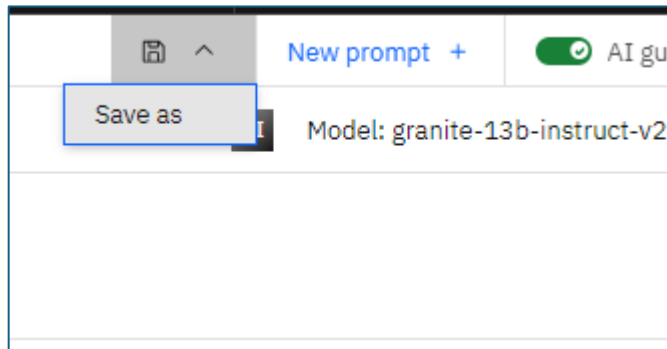
Sample exercises

- [Build an AI stylist with IBM Granite using watsonx.ai](#)
- [Getting Started with Time Series Models on IBM watsonx.ai](#)
- [Build an agentic framework with CrewAI and IBM watsonx.ai](#)

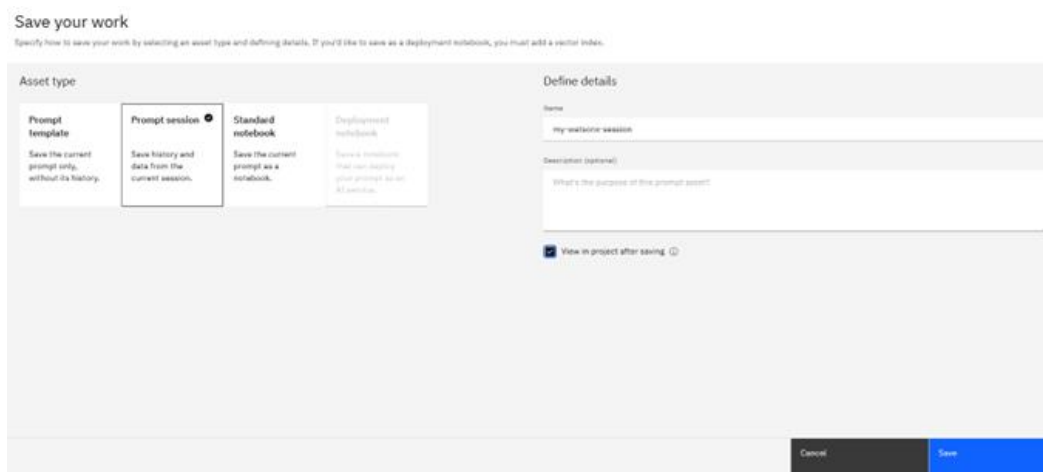
Saving your work

You can save your Prompt Lab editor session for later use.

- At the top of the Prompt Lab screen, select the **Save work** dropdown and choose **Save as**.



- In the **Save your work** widget, select **Prompt session** under **Asset type**, enter a name, check **View in project after saving**, and click **Save**.



IBM Cloud accounts are deactivated at the end of the hackathon. To export your project, open the project's 'Overview' tab, click the 'Export or import project' drop-down beneath the Bell icon, select 'Export project', choose all assets (project sessions cannot be exported), confirm sensitive information has been removed, and click 'Continue export' to download the .zip.

Get started with IBM Open Source

[IBM Granite](#) is a family of AI models purpose-built for business, engineered from the ground up to ensure trust and scalability in AI-driven applications. Granite models deliver strong performance against safety benchmarks and across enterprise tasks from cybersecurity to RAG, and they're available open-source — meaning your team can run them locally or self-host them without provisioning cloud infrastructure.

IBM Granite 3.2 is the latest release in the third generation of Granite models. Headlined by experimental reasoning features and the first official [vision language model \(VLM\)](#), Granite 3.2 introduces several significant new capabilities. Learn more about [Granite 3.2](#).

Granite via Hugging Face and Ollama

Two options for getting Granite running on your own machine:

- [Download Granite on Hugging Face](#)
- [Run locally with Ollama](#)
 - Install Ollama on your local machine before using Granite:
 - [Mac](#)
 - [Windows](#)
 - [Linux](#)
 - [Granite models supported on Ollama](#)

System requirements

To run Granite models locally, it is recommended to use a machine with **at least 32 GB RAM** and a **GPU processor**. Running on lower RAM and CPU is possible but may result in slower performance.

Documentation and learning resources

- [IBM Granite documentation](#) — full reference for [models](#), [agent recipes](#), and [use cases](#).
- [Chat template documentation](#) — recommended for best performance from Granite models.
- [Granite cookbooks](#) — sample use cases for different Granite models.
- [Granite Workshop](#) — hands-on workshop covering text summarization, RAG, time series forecasting, and code generation.

Other Granite quick start samples

- [Build an AI-powered multimodal RAG system with Docling and Granite](#)
- [IBM Granite 3.2: Writing / Debugging code with AI in seconds](#)
- [Build an AI research agent for image analysis with Granite 3.2 Reasoning and Vision models](#)

BeeAI Agentic Framework

[BeeAI](#) is an open-source agentic framework for building production-ready AI agents, optimized for smaller models such as the Granite 3 series.

- [BeeAI Framework for Python](#) — installation and quick example.
- [BeeAI Framework for TypeScript](#) — installation and quick example.
- [Hands-on labs](#) — basics, [Granite-powered ReActAgent](#), and [agent workflows](#).

Get started with IBM Cloudant

[IBM Cloudant](#) is a fully managed JSON document database, built on Apache CouchDB technology and accessible via a simple HTTPS API. Use Cloudant to store user data, application state, prompts and outputs, conversation history, or any structured/unstructured data your hackathon solution generates. It's schema-free, so you can iterate quickly without locking in a data model up front.

Cloudant pairs naturally with Code Engine — deploy your app to Code Engine and use Cloudant as the persistent backend. It also supports full-text search and geospatial indexing if your project needs them.

Pre-configured services: Cloudant for the hackathon is provisioned as a **Lite plan** instance. The Lite plan is free and sufficient for hackathon-scale prototypes — you get 1 GB of data storage, 20 reads/sec, 10 writes/sec, and 5 global queries/sec. If you notice a permission/access issue for any service or the cloud catalog, those services are not required or available for this hackathon.

Access Cloudant

In your IBM Cloud account dashboard, select the **Navigation menu** on the top left of the dashboard and select the **Resource list** option.

Expand the **Databases** section and select the **Cloudant** service instance provisioned for your team. You will be navigated to the Cloudant service instance dashboard, where you can launch the Cloudant dashboard, manage service credentials, and view your plan details.

Get your service credentials

To connect to your Cloudant instance from code, you need a set of service credentials:

- From the Cloudant service instance page, select **Service credentials** in the left sidebar.
- Click **New credential**, give it a name, and click **Add**.
- Expand the new credential to see the URL, API key, and other connection values. Copy and store these securely — your code will use them to authenticate against the Cloudant API.

Getting started

Cloudant works through a simple HTTPS REST API. Some good first steps depending on what you're building:

- [Working with documents](#) — Cloudant stores data as JSON documents. Start here to understand the basic create/read/update/delete pattern.
- [Working with databases](#) — create databases to organize your data.
- [Cloudant Query](#) — declarative query language for filtering documents (similar to MongoDB-style queries).
- [Search indexes](#) — full-text search across documents.
- [Geospatial queries](#) — useful if your solution involves location data.

Documentation and developer tools

- [Cloudant API reference](#) — full HTTPS REST API documentation with code samples.
- [Cloudant SDKs](#) — official client libraries for [Java](#), [Node.js](#), [Python](#), and [Go](#).
- [Getting started with Cloudant](#) — end-to-end walkthrough including connecting from a sample web app.

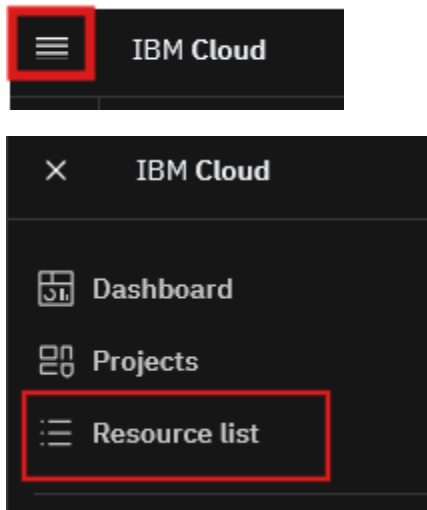
Get started with IBM Code Engine

IBM Code Engine is a fully managed, serverless platform that runs containerized workloads, including web apps, microservices, event-driven functions, and batch jobs. Code Engine can even build container images for you from your source code, so your team can focus on writing code rather than managing infrastructure. Use Code Engine to deploy your hackathon solution as a running app or service that judges and other participants can interact with live.

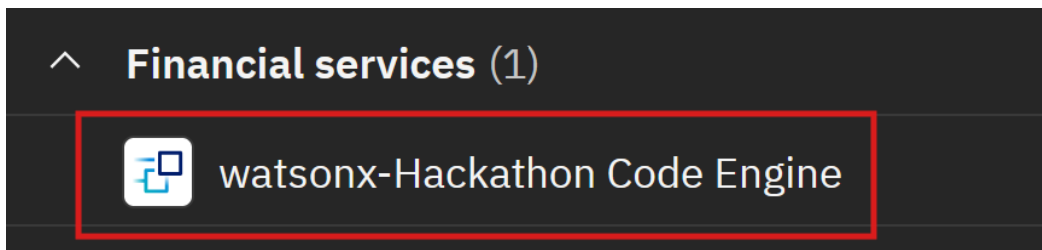
Pre-configured services: Code Engine for the hackathon is pre-configured with only the services required to complete the hackathon. If you notice a permission/access issue for any service or the cloud catalog, those services are not required or available for this hackathon.

Access Code Engine

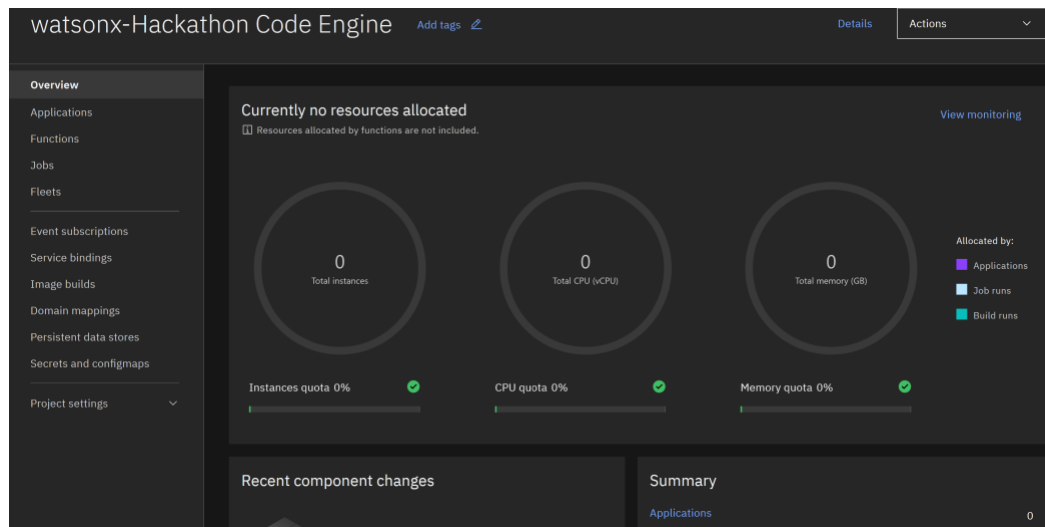
In your IBM Cloud account dashboard, select the **Navigation menu** on the top left of the dashboard and select the **Resource list** option.



Expand the **Financial services** section and select the **watsonx-Hackathon Code Engine** service.



You will be navigated to the **watsonx-Hackathon Code Engine** service instance dashboard.



Getting started

Start by learning the [key terms](#) for Code Engine, then pick one of the following starting points based on what you're building:

- [Deploy an application](#)
- [Create and run a job](#)
- [Run function code](#)
- [Build your first container image from source code](#)
- [Run a fleet](#)

Next steps and developer tools

Learn more about performing these Code Engine tasks from the console or with the [Code Engine CLI](#):

- [Managing projects](#)
- [Deploying applications](#)
- [Running jobs](#)
- [Running functions](#)
- [Building container images](#)

Not sure which type of Code Engine task fits your requirement? See [Planning for Code Engine](#).
For deeper integration:

- [API and SDK reference](#)

- [CLI reference](#)
- [Terraform reference](#)

Quick start hands-on exercises

Try the quick start exercises for sample Code Engine use cases. **Important:** you must use the hackathon-provisioned cloud account to access and use Code Engine to avoid usage issues.

- [Code Engine sample apps and tutorials \(GitHub\)](#)
- [IBM developer tutorials](#)

Appendix: Example use cases

You are not limited to these ideas, but here are several examples of how you could combine the IBM tools above to solve a specific issue within your chosen theme:

Make work less boring

- **Personalize learning and skill development:** Facilitate self-guided learning by using AI as an intelligent tutor or coach. Create tailored learning paths and recommend relevant courses, articles, and tools to help employees acquire new skills.
- **Facilitate interactive collaboration and brainstorming:** Use AI as a smart collaborator during brainstorming sessions to generate fresh ideas, challenge assumptions, provide problem-solving suggestions, and synthesize input from team members.
- **Enable intuitive communications:** Automate and improve communication tasks such as email responses, reminders, scheduling, and interactions with clients or teammates.
- **Generate intelligent content for creative projects:** Assist with content creation by helping design or generate draft text, marketing materials, reports, or presentations.
- **Automate and optimize personal tasks and workflows:** Automate routine tasks like data entry, report generation, and scheduling, freeing up time for more meaningful, engaging, and creative work.